

Course Syllabus

General Information

Faculty	Information Technology		
Department	Cybersecurity Dept.		
Semester	1 st	Academic Year	2025/2026

Course Details

Course Title	Ethical Hacking & Penetration Testing			Course No.	0453403		
Type of Learning	☒ On- campus ☐ Blended (..... Platform) ☐ Online (..... Platform)						
Credit Hours	3	Theoretical	3	Practical	0		
Course Level (according to JNQF)			7	JNQF Hours*		12.15	
Pre-requisite	0452302- Network Security & Defense			Co-requisite			

Course Instructor Information

Instructor Name	Dr. Ahmad Alnajjar		
Instructor E-mail	anajar@meu.edu.jo		
Course Time	8-9 Sun, Tues, Thru		
Office Hours	9-10 Sun. Mon. Tue, Wed, Thru		
Office Number	N134	Office Phone	--
Lab Instructor Name –If assigned-	--		

Sources and References

1. Textbook
1. Ethical Hacking Essentials eBookV1 , EC Council, 2022,
2. References
CEH v12 Certified Ethical Hacker Study Guide, <u>Ric Messier</u> ,SYBEX, 2023, ISBN: 978-1394186921

Teaching Methods

1. Lectures	2. Questions	3. Reports	4. Exercises
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Course Description

Focus on the concepts and techniques used to hack networks and systems. The common techniques such as Vulnerability Assessment, Social Engineering, Network Level Attacks, and Web Application Attacks are introduced. For each item, students will be trained using case studies to acquire the needed skills.

Course Learning Outcomes (CLOs) and its correlation with Program Learning Outcomes (PLOs) and JNQF Descriptors

No.	CLO's	PLO's	JNQF Descriptors		
			Knowledge	Skill	Competency
1-	Understand The deferent concepts of Ethical hacking (25%)	1	K1-I		
2-	Practice different techniques password cracking and Countermeasures (20%)	5			C2-P
3--	Practice different techniques for Network Level Attacks and Countermeasures (30%)	5			C2-P
4-	Practice different techniques Social Engineering Attacks and Countermeasures (10%)	5			C2-P
5	Practice different techniques for Web Application Attacks and Countermeasures (15%)	5			C2-P

Course Topics

	Topics	Lectures		CLOs	Assessment	
		Lecture No.	Lecture Type		Assessment Tool	Summative / Formative
Week 1	Ethical hacking Fundamentals What is Hacking What is Ethical hacking Hacking Cycle Cyber kill chain methodology	Lecture (1)	Lecture On-campus	1	Discussion & Questions	Formative
		Lecture (2)	Lecture On-campus	1		
		Lecture (3)	Lecture On-campus	1		
Week 2	Reconnaissance	Lecture (1)	Lecture On-campus	1	Discussion & Questions	Formative
		Lecture (2)	Lecture On-campus	1		
		Lecture (3)	Lecture On-campus	1		
Week 3	Scanning and Enumeration	Lecture (1)	Lecture On-campus	1	Discussion & Questions	Formative
		Lecture (2)	Lecture On-campus	2		

		Lecture (3)	Lecture On-campus	2		
Week 4	Threads and Malware	Lecture (1)	Lecture On-campus	1	Discussion & Questions Assignment	Summative
		Lecture (2)	Lecture On-campus	2		
		Lecture (3)	Lecture On-campus	2		
Week 5	Vulnerabilities Assessment	Lecture (1)	Lecture On-campus	1	Discussion & Questions	Formative
		Lecture (2)	Lecture On-campus	1		
		Lecture (3)	Lecture On-campus	2		
Week 6	Password cracking: part 1	Lecture (1)	Lecture On-campus	2	Discussion & Questions Quiz	Summative
		Lecture (2)	Lecture On-campus	2		
		Lecture (3)	Lecture On-campus	2		
Week 7	Password Cracking:Part2	Lecture (1)	Lecture On-campus	2	Discussion & Questions	Formative
		Lecture (2)	Lecture On-campus	2		
		Lecture (3)	Lecture On-campus	3		
Week 8	Midterm					Summative
Week 9	Password Cracking tools	Lecture (1)	Lecture On-campus	2	Discussion & Questions	Formative
		Lecture (2)	Lecture On-campus	2		
		Lecture (3)	Lecture On-campus	3		
Week 10	Sniffing	Lecture (1)	Lecture On-campus	3	Discussion & Questions	Formative
		Lecture (2)	Lecture On-campus	3		
		Lecture (3)	Lecture On-campus	3		
Week 11	Network Attacks: part 1	Lecture (1)	Lecture On-campus	3	Discussion & Questions Quiz	Summative

		Lecture (2)	Lecture On-campus	3		
		Lecture (3)	Lecture On-campus	3		
Week 12	Network attacks: part 2	Lecture (1)	Lecture On-campus	3	Discussion & Questions	Formative
		Lecture (2)	Lecture On-campus	3		
		Lecture (3)	Lecture On-campus	4		
Week 13	Social Engineering	Lecture (1)	Lecture On-campus	4	Discussion & Questions Assignment	Summative
		Lecture (2)	Lecture On-campus	4		
		Lecture (3)	Lecture On-campus	4		
Week 14	Web Server attacks	Lecture (1)	Lecture On-campus	5	Discussion & Questions	Formative
		Lecture (2)	Lecture On-campus	5		
		Lecture (3)	Lecture On-campus	5		
Week 15	Project Discussion	-	Lecture On-campus	5	Discussion & Questions	Summative
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Week 16	Final Exam	-	Lecture On-campus	1-5	Discussion & Questions	Summative

Assessment Tool	Weight %	Expected Due Date
Mid Exam	25%	Week 8
Assignments	10%	Week 4, 13
Quizzes	10%	Week 6,11
Project	15%	Week 15
Final Exam	40%	Week 16

Course Policies

Item	Policy
Quizzes	- No makeup quizzes. - A minimum 2 quizzes are given. - Each Quiz is out of 5. - If five quizzes or more are given then the lowest quiz's grade is dropped.
Exams	The format of the exams is generally (but NOT always) as follows: General Definitions, Multiple-Choice, True/False, Analyze a Problem, Essay Questions, etc.
Makeup Exams	Makeup exam should not be given unless there is a valid excuse.
Drop Date	Last day to drop the course is according to the University regulations.
Academic Honesty	- Cheating or copying on exam or quiz is an illegal and unethical activity. - Standard MEU policy will be applied. - All graded assignments must be your own work (your own words).
Attendance	- Excellent attendance is expected. - MEU policy requires the faculty member to assign ZERO grade if a student misses 20% of the classes. - Sign-in sheets will be circulated. - If you miss class, it is your responsibility to find out about any announcements or assignments you may have missed.
E-Learning	It is your responsibility to check the course's eLearning website regularly (at least once every 2-3 days) for announcements and material.
Workload	The student should expect to spend 6 hours per week as an average workload.
Graded Exams	Instructor should return exam papers graded to students within one week after the exam date.
Participation	- Participation in and contribution to class discussions will affect your final grade positively. Raise your hand if you have any questions. - Making any kind of disruption and (side talks) in the class will affect you negatively.
Assignment/ Projects	- Assignments are as mentioned in the previous table. - Students organize themselves in teams of (2-3) students each. Each team will give at least one presentation regarding a selected case, delivering a fully working application. - More details about phases and topics of the projects will be given later on.
Others	

Rubrics and Feedbacks (If Applicable)

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*** JNQF Hours Calculation**

Activity	Duration/ week	Times/ Semester	No. of Hours
Learning Activities			
Lectures and Seminars	3 hours	15	45
Tutorials	0	0	0
laboratory	0	0	0
Activities for Assessment and Evaluation			
First Exam/ Midterm	1.5 hours	1	1.5
Second Exam (If applicable)	0	0	0
Final Exam	2 hours	1	2
Quizzes	0	0	0
Self-Learning Activities			
Assignments/ Homework	1.35 hours	4	5.5
Case Studies	0	0	0
Presentations	0	0	0
E-learning	0	0	0
Extra Readings	4.5	15	67.5
Total Hours	121.5		
JNQF Hours	=121.5/10 = 12.15		